Modeling Resource Management in Concurrent Computing Systems

Isaac D. Scherson (aka The Schark)

Dept. of Computer Science (Systems)

School of Information and Computer Science

University of California, Irvine

Irvine, CA 92697-3425

isaac@ics.uci.edu - www.ics.uci.edu/ ˜ isaac

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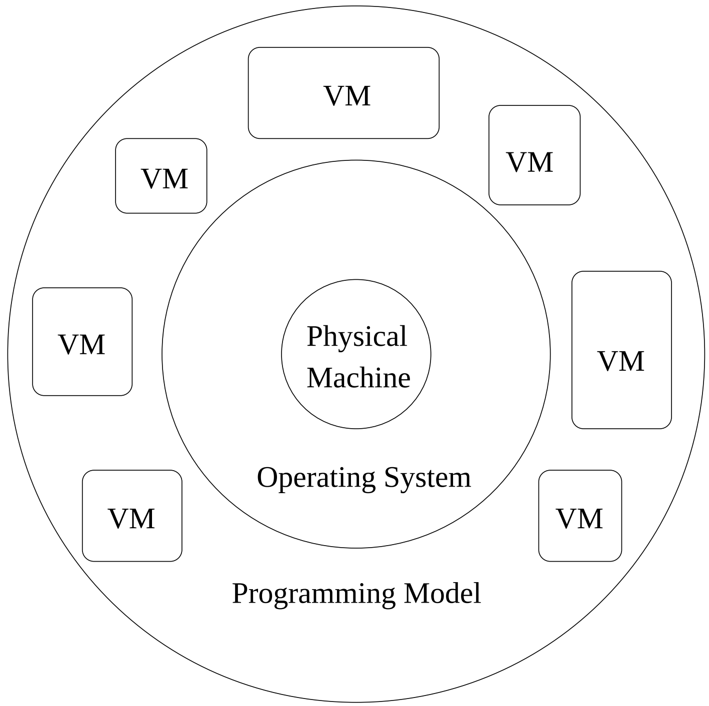
Warning !

This is a thought-provoking presentation. The ideas are very simple and are presented to motivate a little formal thinking about the scheduling problem in Concurrent Computing Systems.

Concurrent Computing OS Model



VM - Virtual Machine

Each VM is a user job

General Purpose Parallel Operating System

The Programming Model

- There seems to be a convergence on the data parallel programming in the form of HPF and Fortran 90.
- In a survey of 120 parallel algorithms from three ACM Symposia on Theory of Computing (STOC), all were found to be data parallel.
- This preponderance of the data parallel programming model is perhaps because it allows one to express a large degree of parallelism, while retaining the single-thread-of-control philosophy of sequential programming.

The Machine Model

- A natural way of executing a data parallel program is on a SIMD machine.
- BUT . . . data parallelism does not equal SIMD.
- A data parallel program may be executed on an asynchronous MIMD machine:
  - Chief advantage: It is possible to run several jobs on a MIMD machine simultaneously.
  - A MIMD machine does not force unnecessary synchronization after every instruction, or unnecessary sequentialization of non-interfering branches, as a SIMD machine does.

SPMD Programs

- The combination of the data parallel programming model and a MIMD machine model is called a SPMD execution model
- SPMD stands for Single Program Multiple Data
  - All processors execute the same program, but may be at different stages of the program at a given time
- For the remainder of this presentation, our discussion is predicated on the SPMD execution model.

Virtual Processors

- A program is expressed in a language that describes a virtual machine.
- For a data parallel program, the virtual machine consists of a (typically large) number of identical *virtual processors* (VPs), communicating through an interconnection network.
  - For instance, the standard data parallel program to multiply two $N \times N$ matrices may be viewed as a virtual machine consisting of N^2 VPs communicating in a mesh.

Virtual Processors (Cont'd)

- For many years now, the concept of virtual processors has been relegated to the status of a mere logical aid to programmers
  - In our view, the notion of virtual processors should form the fundamental basis for the definition of a Concurrent Computer System and for the resource management strategies embedded in the OS.

Redefining Massive Parallelism

- A Programmer sees only a VM with as many Virtual Processors as s/he needs.
- Solutions are cast in the space of the problem and not on that of the physical machine.
- The measure of Massive Parallelism is through the number of VPs the physical machine is capable of emulating per unit of time.
- HENCE . . . It does not matter what the physical machine looks like, provided it can emulate Concurrent Computing Virtual Machines.

Resource Management Operations

- **Spatial scheduling:** A space sharing policy that defines which physical processor each VP is allocated to.
- **Temporal scheduling:** A time sharing policy that dictates how each physical processor switches between the execution of the VPs allocated to it.
- **Load balancing:** Static and/or Dynamic redistribution of VPs among physical processors responding to some predetermined objective function.

Resource Management Operations

- **Memory and I/O problems:** Memory limitations and I/O bottlenecks may also be phrased in terms of VPs. Memory limitations occur when a processor does not have enough local memory to hold all the VPs allocated to it. I/O bottlenecks are most pronounced while loading (roll-in) the VPs of a job from the disk into the local memories of various processors.

AND NOW ...

A game you can play in your free time . . . . . .

Model of the System

- Let $\Pi = \{\pi_1, \pi_2, \ldots, \pi_n\}$. denote the set of physical processors or processing elements (PEs).
- Interconnection network details are ignored, only assuming that it *guarantees* the completion of all communications within a reasonable time bound (not necessarily a constant, as in the PRAM model).
- Let $\mathcal{J} = \{J_1, J_2, \ldots, J_m\}$ denote the set of jobs in the system at some time.
- For notational convenience, job J_i is identified with its set of VPs $\{P_{i1}, P_{i2}, \ldots, P_{i|J_i|}\}$.

Handling Time

- Break time into **time slices**: discrete intervals of equal length.
- The time slice is assumed to be machine dependent and constant.
- Within each time slice, every π_i runs some VP chosen from the VPs allocated to it, or it idles.

Handling Time (Cont'd)

- A PE π_i runs a VP of a Job until:
  - The VP issues a communication request which cannot be completed (data to be read is not ready), or
  - The time slice expires.
- The communication instruction is considered an indivisible instruction and time slices are extended accordingly.

Execution of Programs

- *For simplicity of analysis,* it is assumed that jobs run indefinitely.
  - The OS schedules an arriving Job as if the job is going to last forever in the system
- This approximation reflects that the time slice is very small (in the order of milliseconds) compared to the execution times of the jobs (in the order of minutes).
- The assumption of infinite running time might be removed with negligible change to the final results, but with significant complications in the definitions and proofs.

The 2 Types of Scheduling

- Given a job, J, as a set of $\left|J\right|$ VPs, how is this job allocated over the n physical PEs of the concurrent computer so that the resources are efficiently utilized?
- More generally, how are VPs of the same and different jobs, say $J_1, J_2, \ldots, J_m$, allocated over the PEs to efficiently utilize all system resources?
  - This is called the problem of spatial scheduling or spatial allocation.
- Intuitively. Spatial schedule: "Where to run the VPs of job J?". Temporal schedule: "When to run the VPs of job J?".

...and more formally

Definition 1 *A* **spatial schedule** *is an n-tuple allocation function (The terms* **allocation** *and* **spatial scheduling** *will be used interchangeably)*

$$\mathcal{A} : \mathcal{J} \longrightarrow I\!N^{|\Pi|}$$

such that for any job J, $\mathcal{A}(J) = (a_1, a_2, \ldots, a_n)$, a_i VPs of J are allocated to processor $\pi_i \in \Pi$, ($\sum_{i=1}^{n} a_i = |J|$).

. . . and more formally (Cont'd)

Definition 2 *A* **temporal schedule for a time slice** t *is defined as an n-tuple function, $S(t)$, such that $S(t)[i]$ is the job run by π_i in the time slice t. A* **temporal schedule**, *$\mathcal{S}$, is a time ordered sequence of $S(t)$ for all $t > 0$.*

Definition 3 *Define a schedule, $\mathcal{S}$, to be* **impartial** *if every job $J \in \mathcal{J}$ occurs infinitely many times in $\mathcal{S}$.*

An impartial schedule ensures that every job in it completes.

Example

Consider a machine with 5 PEs and 5 jobs:

J_1 requires 1 VP,

J_2 requires 4 VPs,

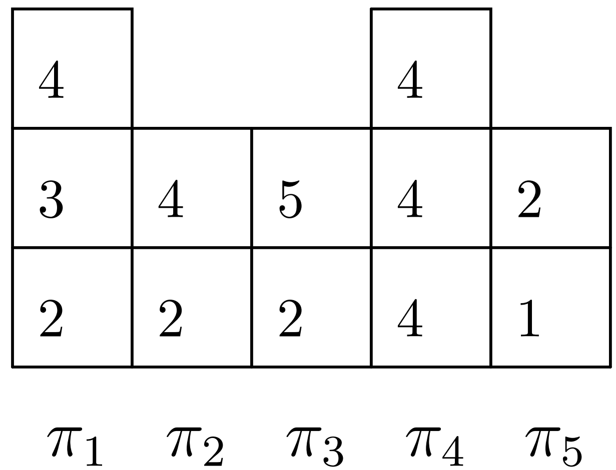
J_3 requires 1 VP,

J_4 requires 5 VPs,

J_5 requires 1 VP.

... suppose we allocate the VPs of the Jobs as follows ...

Example (Graphical)



With the following allocation functions:

$$\mathcal{A}(J_1) = (0, 0, 0, 0, 1) \qquad \mathcal{A}(J_2) = (1, 1, 1, 0, 1)$$

$$\mathcal{A}(J_3) = (1, 0, 0, 0, 0) \qquad \mathcal{A}(J_4) = (1, 1, 0, 3, 0)$$

$$\mathcal{A}(J_5) = (0, 0, 1, 0, 0)$$

Job Execution

- **Global communication:** all VPs of a given Job execute a communication instruction.
- **Legal execution of a Job:** no VP of the Job is ahead of any other VP of the same job by more than one global communication step at any time.
- **Legal scheduling strategy:** will always choose any trailing VP before any non-trailing VP.
- A **step** begins when all VPs of a job have executed the same number of global communications and it ends at the occurrence of the same condition at a future time slice (after at least one VP of the job has been executed).

A Job's Schedule Vector

Definition 4 *Define a job J's t-**schedule vector**, $\epsilon(t, J)$ as a 0-1 n-tuple such that,*

$$\epsilon(t, J)[k] = \begin{cases} 1 & \textit{if } \pi_k \in \Pi \textit{ runs job } J \textit{ in time slice } t \\ 0 & \textit{otherwise} \end{cases}$$

A Job's Progress Vector

Definition 5 *Define* **progress of a job** *(at the begining of a time slice t) as an n-tuple function*

$$\mathcal{P} : t \times \mathcal{J} \longrightarrow \mathbb{N}^{\Pi}$$

such that:

1. $\mathcal{P}(t, J) = \vec{0}$ *for* $t = 0$,
2. *If* $\mathcal{P}(t-1, J) + \epsilon(t, J) = \mathcal{A}(J)$ *then* $\mathcal{P}(t, J) = \vec{0}$, *else* $\mathcal{P}(t, J) = \mathcal{P}(t-1, J) + \epsilon(t, J)$ *(where the addition is componentwise)*
3. $\mathcal{P}(t, J) \leq \mathcal{A}(J)$ *where the* $\leq$ *order is applied componentwise.*

Progress Vector in English

The intuition behind the progress vector is as follows:

Start at the first step of a job J, let job J proceed on the processor's having 1's on the $\epsilon(t, J)$. If the progress vector reaches $\mathcal{A}(J)$ then another step of job J is complete and progress counters associated with each processor are reset. A new step cannot be started until the previous one is complete. Every job has a progress vector associated with it at the beginning of every time slice.

A more complete Example

Consider an allocation on 5 processors (same example):

4			4	
3	4	5	4	2
2	2	2	4	1
π_1	π_2	π_3	π_4	π_5

And consider the problem of scheduling J_4.

A more complete Example (Cont'd)

An arbitrary schedule for job J_4 is as follows:

$$\begin{aligned}\mathcal{A}(J_4) &= (1,1,0,3,0)\\ P(0,J_4) &= (0,0,0,0,0)\\ \epsilon(1,J_4) &= (0,1,0,1,0)\end{aligned}$$

$$\left[\begin{array}{rclcl} P(0,J_4)+\epsilon(1,J_4) &=& P(1,J_4) &=& (0,1,0,1,0)\\ && \epsilon(2,J_4) &=& (1,0,0,1,0)\end{array}\right.$$

$\epsilon(1,J_4) = (0,1,0,1,0)$ indicates that processors π_2 and π_4 run job J_4. $\epsilon(1,J_4)$ is added to $P(0,J_4)$ componentwise to give the new progress vector $P(1,J_4)$ for job J_4 and so on.

A more complete Example (Cont'd)

Continuing to schedule J_4 ...

$$\left[\begin{array}{rcl} P(1, J_4) + \epsilon(2, J_4) & = & P(2, J_4) & = & (1, 1, 0, 2, 0) \\ & & \epsilon(3, J_4) & = & (0, 0, 0, 1, 0) \end{array} \right.$$

$$P(2, J_4) + \epsilon(3, J_4) = (1, 1, 0, 3, 0)$$

Now, $P(3, J_4) = \mathcal{A}(J_4)$, so reset, $P(3, J_4) = (0, 0, 0, 0, 0)$

System's Progress and Schedule

Definition 6 *Define the* **system progress matrix**, Q, *as a $(n \times m)$-array of progress vectors (columns) of all jobs (rows) in the system.*

Definition 7 *Define the* **system schedule matrix**, S, *as a $n - column$ matrix such that each column corresponds to a physical processor while each row corresponds to a time slice. Each entry is numbered with the job number whose VP is executed at that time slice.*

A Schedule for the Example

Time/PE	π_1	π_2	π_3	π_4	π_5
1	2	2	2	4	1
2	3	4	5	4	2
3	4			4	

The same three lines can be repeated until all jobs terminate.

The above will be dubbed the **Scheduling Matrix**

HINT: We'll be seeing periodic schedules.

Another Schedule for the Example

Time/PE	π_1	π_2	π_3	π_4	π_5
1	2	2	2	4	2
2	3	4	5	4	1
3	4		5	4	1
4	2	2	2	4	2
5	3	4	5	4	1
6	4		5	4	1

Same period as the previous schedule (3) but only one idle slice per period.

Schedules and Allocations

- A Schedule will strongly depend upon the initial Allocation.
- Challenge: Find another allocation/schedule for the example such that there are no idle time slices.

Metrics

There are two points of view when measuring the system's performance.

- One focuses on the functional quality from the system manager's point of view, who is concerned with the throughput and utilization of the machine.
- The second focuses on the functional quality from the users' point of view who expects the system to respond within a specific time.

Scheduling Metrics

For the case of Scheduling:

- Question: Which schedule is the “best”?
- The parallel OS should be able to satisfy two objectives:
  - Minimize processor idling time (important for the System).
  - Guarantee an upper limit on the system response time to every job (important to the single user).

Idling Ratio

Definition 8 *For a n-processor system, for a schedule $\mathcal{S}$, and over and interval of Δt time slices, define an* **idling ratio** *or* **throughput** $\imath_{\mathcal{S}} : \mathbb{N} \rightarrow \mathbb{R}$ *as:*

$$\imath_{\mathcal{S}}(\Delta t) = \frac{\sum_{k \in \Delta t} i_k}{n \times \Delta t}$$

where i_k is the total number of idling processors during time slice k.

The idling ratio is a measure of the resource utilization.

Idling Ratio for the Example

Time/PE	π_1	π_2	π_3	π_4	π_5
1	2	2	2	4	1
2	3	4	5	4	2
3	4			4	

Idling Ratio for this schedule over the period of 3 time slices =

$$3/15 = 1/5 = 20\%$$

Idling Ratio for the Example

Time/PE	π_1	π_2	π_3	π_4	π_5
1	2	2	2	4	2
2	3	4	5	4	1
3	4		5	4	1

Idling Ratio for this schedule over the period of 3 time slices = $1/15$

An Interesting Case

Consider a system where only one job is running and utilizes all processors (J_1 has 5 VPs):

Time/PE	π_1	π_2	π_3	π_4	π_5
1	1	1	1	1	1

An Interesting Case

. . . SUDDENLY . . .

An Interesting Case

. . . SUDDENLY . . .

Another job J_2 arrives and requires 4 VPs

An Interesting Case

In a hurry, the system allocates and schedules J_2 in the apparently obvious manner:

Time/PE	π_1	π_2	π_3	π_4	π_5
1	1	1	1	1	1
2		2	2	2	2

With an Idling Ratio of $1/10 = 10\%$

An Interesting Case

BUT . . . Consider the following 0% Idling Ratio Schedule:

Time/PE	π_1	π_2	π_3	π_4	π_5
1	1	2	2	2	2
2	1	2	2	2	2
3	1	2	2	2	2
4	1	2	2	2	2
5	1	2	2	2	2

The response time of J_1 was sacrificed in favor of full 100% Physical Processor utilization. AN UNHAPPY USER !!!

Happiness Function

Definition 9 *Define the* **happiness function**, $\hbar_{\Delta t}(J)$, *of a job* J *for a time length* Δt *as,*

$$\hbar_{\Delta t}(J) = \min_{t \geq 0} \left\{ \frac{\sum_{\tau=t}^{t+\Delta t-1} \sum_{\pi \in \Pi} \epsilon(\tau, J)}{|J| \cdot \Delta t} \right\}$$

The time interval Δt is the **Impatience Latency** of the user.

Happiness Function

- Intuitively, happiness represents the machine power that a job is given during a time slice. The term in the braces represents the fraction of the full parallel speedup that was achieved by job J in time interval $(t, t + \Delta t - 1)$.
- Δt represents the expected **impatience latency** of the user. It is the minimum time the user is required to wait before the system responds.

Virtual Happiness

- The proposed parallel OS paradigm allows the user to program in a virtual parallel model with no limit on the number of VPs in a job.
- To get a handle on the maximum parallel speedup that can be achieved for a job given n physical processors, define **virtual happiness**,
  $v(J) = \min\{1, \frac{n}{|J|}\}$.
- Virtual happiness of a job J represents the happiness of the job if the entire machine is used by J.
- Hence, J cannot be happier than $v(J)$.

Happiness is a Balancing Act

- A Job (user) will be happier if it acquires more parallel speedup over time.
- If a schedule is unable to provide the required happiness for a job, rescheduling with a different allocation may have to be considered.
- If the required happiness for a set of jobs cannot be achieved by any allocation, then some jobs must be queued for later allocation.
- A happy schedule is not necessarily optimal for throughput. It merely guarantees an upper limit on the response time.

Periodic Temporal Schedules

Definition 10 *For an allocation $\mathcal{A}$,*
$\mathcal{S}_p = S(t), S(t+1), \ldots, S(t+p-1)$ *is called a* **periodic schedule***, iff* $\exists\, p > 0$ *such that* $S(t) = S(t+p)$ *for all* $t > t_s$*, where* t_s *is called the startup time of the schedule.* p *is the period of the schedule.*

Also for each job J in $\mathcal{S}_p$,

$$\sum_{1 \le t \le p, 1 \le i \le |\Pi|} [S(t)[i] = J] = k \cdot |J|$$

must be true for some positive integer k to ensure that each job executes iat least k steps during each period (k need not be the same for every job).

How far did we go?

- This is *Work in Progress*. A lot to be done !
- We have proven a few theorems on Impartial Periodic Schedules.
  - Periodic Schedules are the BEST !!
  - We have conditions to minimize Idling Ratio and maximize Happiness.

Collaborations Welcome

Anybody interested ? . . . Let's work together !!!

THANKS !!